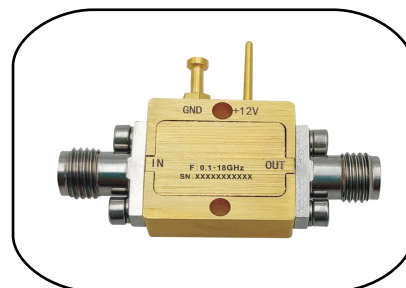




Ultra Wide Band Low Noise Amplifier 0.1GHz ~ 18GHz

Features

- Gain 15dB Typical
- P1dB Output Power 15 dBm Typical
- Supply Voltage +12VDC
- 50 Ohm Matched Input/Output
- Low Noise Figure +1.8dB Typical
- Gain Flatness +/-1.5dB



Typical Applications

- Wireless Infrastructure
- 5G communication
- Test and measurement Instrument

RF Microwave & VSAT
Fiber Optics

Parameter	Min.	Typ.	Max.	Units
Frequency Range	0.1		18	GHz
Gain	13.5	15		dB
Gain Flatness		± 1.5		dB
Gain Variation Over Temperature (-40°C~+85°C)		± 1.5		dB
Noise Figure		1.8	2.8	dB
Input VSWR		1.5	2.0	:1
Output VSWR		1.5	2.0	:1
Output 1dB Compression Point (P1dB)	12	15		dBm
Saturated Output Power (Psat)		17		dBm
Output Third Order Intercept (OIP3)		25		dBm
Supply Current (Vcc=+12V)		35	60	mA
Isolation S12		-20		dB

Weight	Net	0.35 Max .ounces	Impedance	50ohms
	Including Heat Sink	0.8 Max .ounces		
Input / Output Connectors		SMA-Female	Material	Aluminum
Finish	Gold Plated	Package Sealing	Epoxy Sealed (Standard)	
			Hermetically Sealed (Option with extra charge)	

WZ-47, BASEMENT, BUDDHELLA VILAGE VIKAS PURI NEW DELHI - 110018

PH:- 011-28533349, MOB:- 91-7217885948, FAX:- 011-28533349

EMAIL:- INFO@SYNERGYTPL.COM, WEBSITE:- WWW.RFCONNECTOR.IN



Ultra Wide Band Low Noise Amplifier 0.1GHz ~ 18GHz

Absolute Maximum Ratings

Operating Voltage	+15V
RF Input Power (RFIN)	+20dBm

Biasing Up Procedure

Step 1	Connect Ground Pin
Step 2	Connect input and output
Step 3	Connect +12V biasing

Power OFF Procedure

Step 1	Turn off +12V biasing
Step 2	Remove RF connection
Step 3	Remove Ground

Environmental Specifications

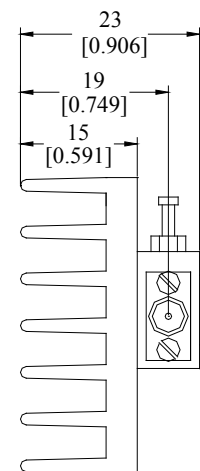
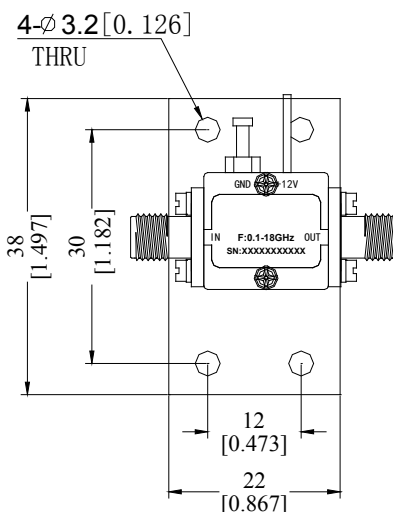
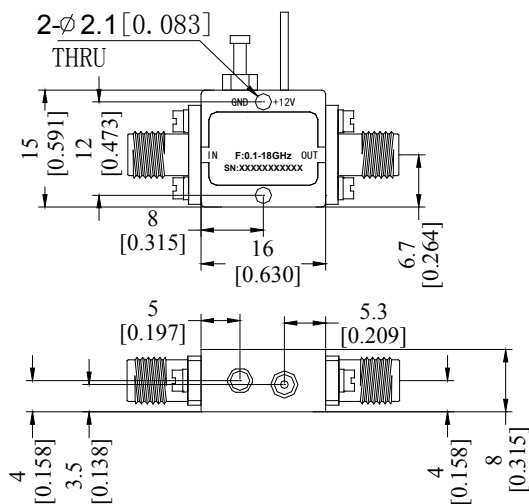
Operational Temperature	-40°C~+85°C(Case Temperature)
Storage Temperature	-50°C~+105°C
Altitude	30,000 ft. (Epoxy Sealed Controlled environment)
	60,000 ft. 1.0psi min (Hermetically Sealed Un-controlled environment) (Optional)
Vibration	25g RMS (15 degrees 2KHz) endurance, 1 hour per axis
Humidity	100% RH at 35°C, 95%RH at 40°C
Shock	20G for 11msec half sine wave, 3 axis both directions

Outline Drawing:

All Dimensions in mm (inches)

Tolerances $\pm 0.1(0.004)$ (Excl heatsink)

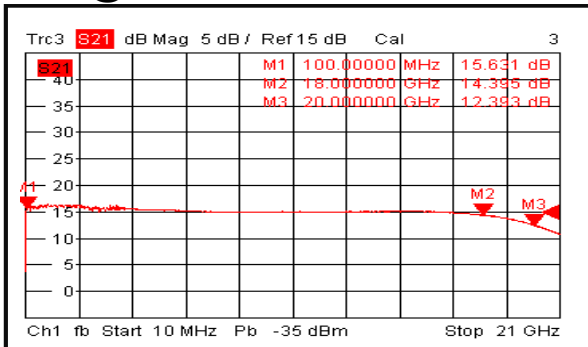
Heat Sink required during operation(Sold Separately)



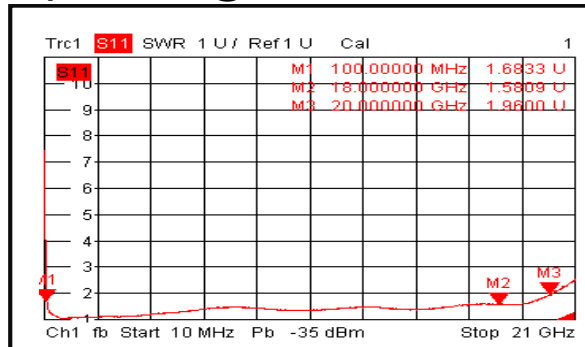


Ultra Wide Band Low Noise Amplifier 0.1GHz ~ 18GHz

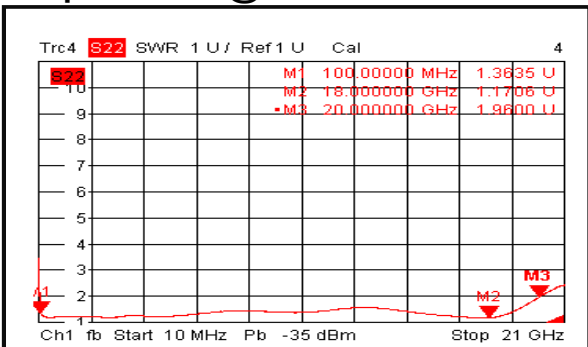
Gain @+25°C



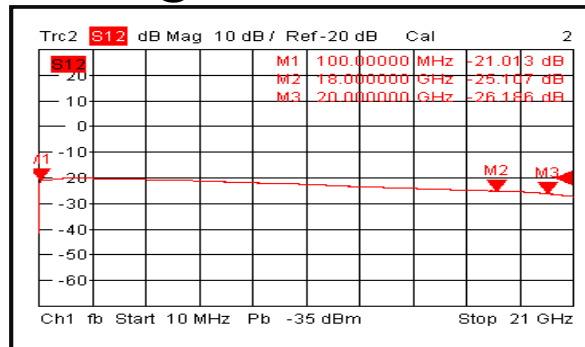
Input VSWR @+25°C



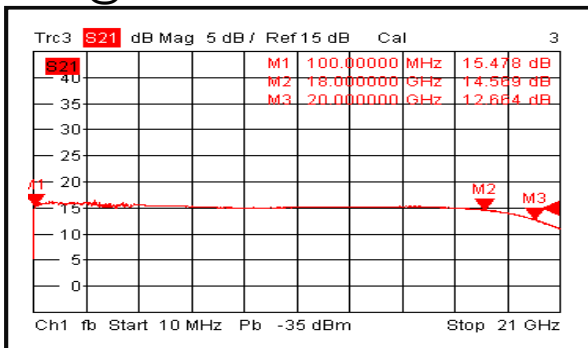
Output VSWR @+25°C



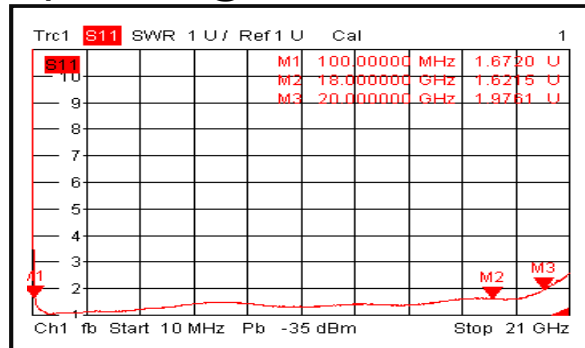
Isolation @+25°C



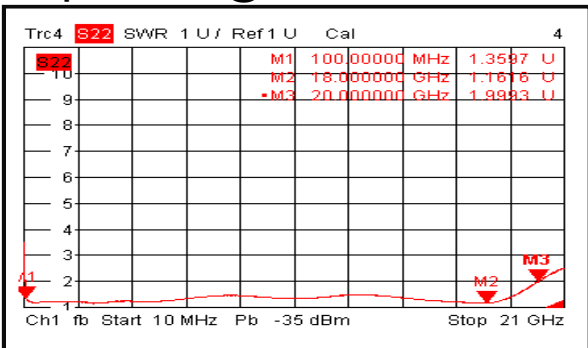
Gain @-40°C



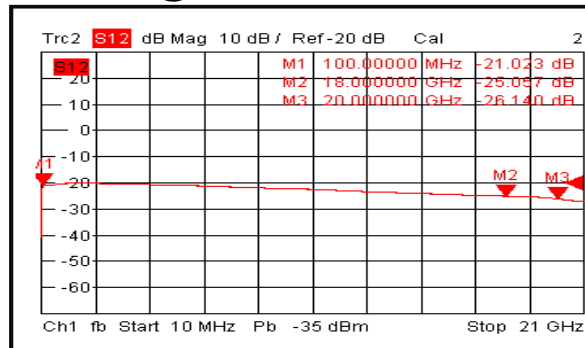
Input VSWR @-40°C



Output VSWR @-40°C



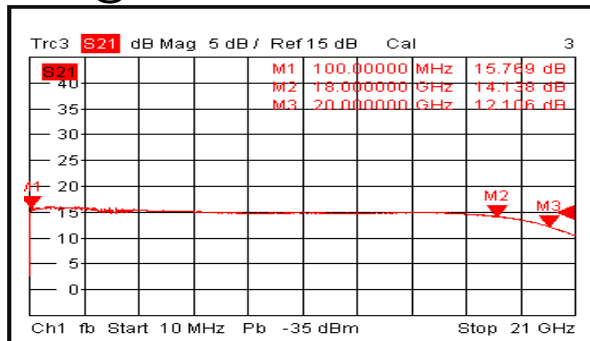
Isolation @-40°C



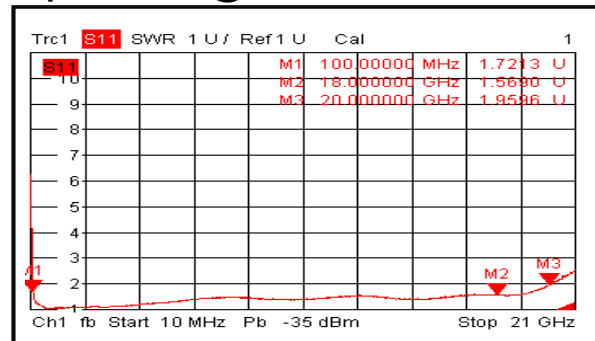


Ultra Wide Band Low Noise Amplifier 0.1GHz ~ 18GHz

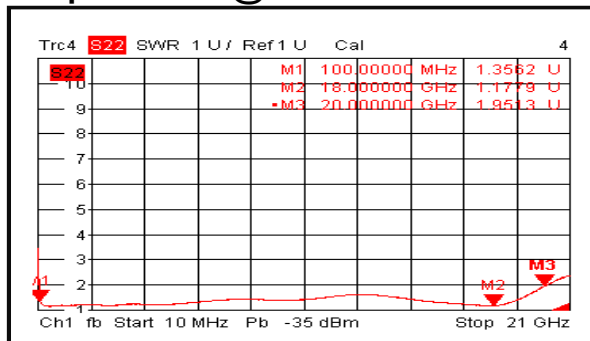
Gain @+85°C



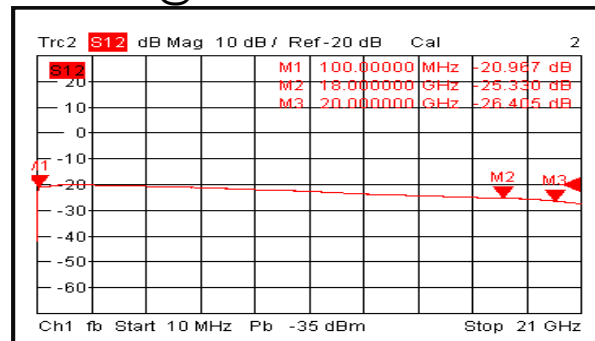
Input VSWR @+85°C



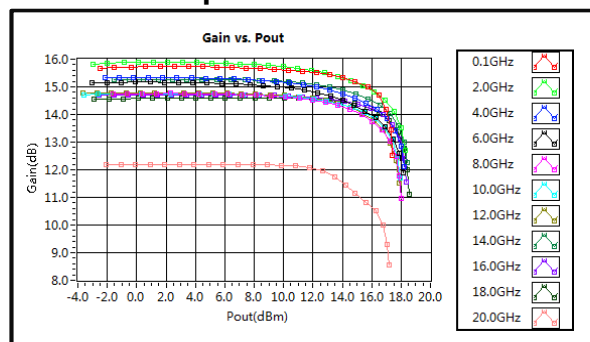
Output VSWR @+85°C



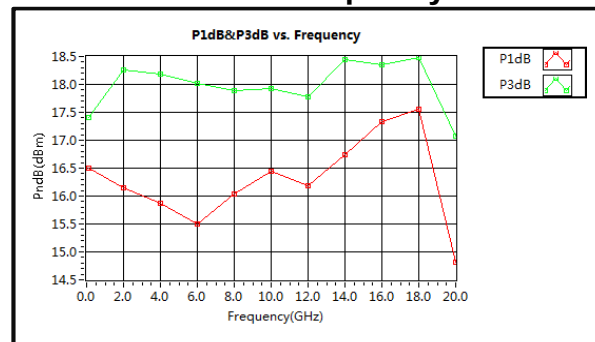
Isolation @+85°C



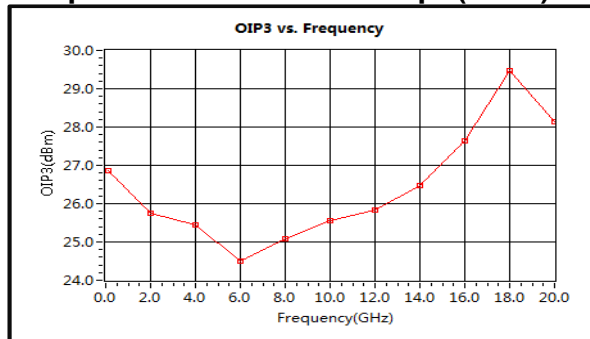
Gain vs. Output Power



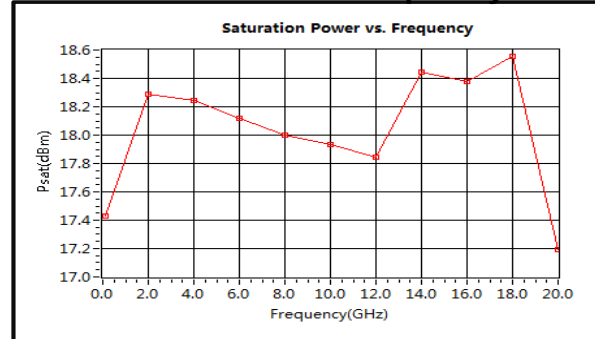
P1dB & P3dB vs. Frequency



Output Third Order Intercept (OIP3)



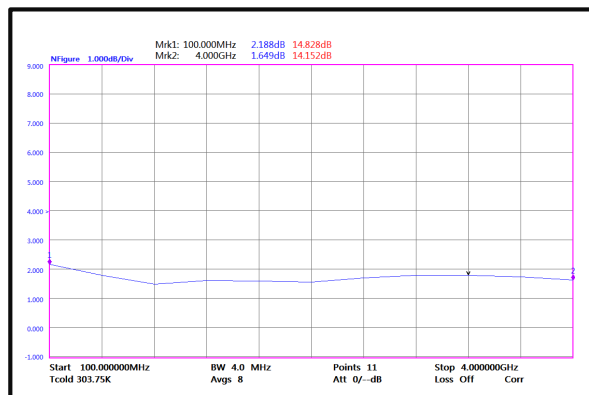
Saturation Power vs. Frequency



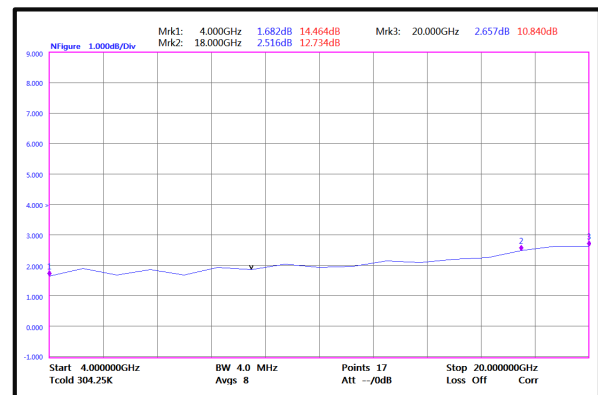


Ultra Wide Band Low Noise Amplifier 0.1GHz ~ 18GHz

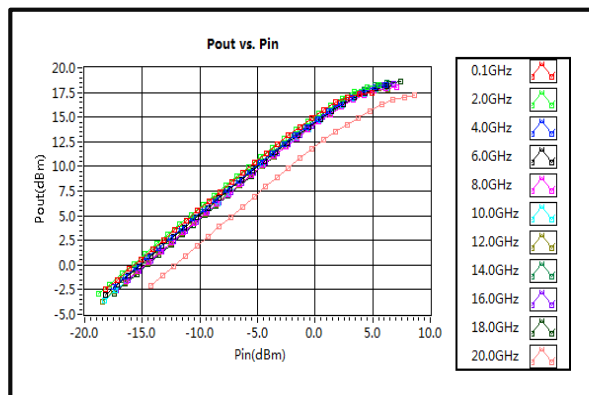
Noise Figure(0.1-4GHz)



Noise Figure(4-20GHz)



Pout vs. Pin



Current vs. Pout

